

INSERT COIN • CS 1 • 30 MIN

LEVEL 2

LOOPS & CONDITIONALS

Block-based programming – read the shape, then build it

`loop` = el bucle

`condition` = la condición

`block` = el bloque

Stand Up

Answer these 3 questions with your group

1. What is your name?
2. What was something that you learned during last class?
3. What was an obstacle you faced while playing last class game?



Agenda:

1. Stand up
2. loops & conditionals
3. Worksheet
4. Game

02 Why Loops: 10 Blocks vs. 1

Same result on screen. Very different code. / Mismo resultado, código muy distinto.

REDUNDANT / REDUNDANTE

move 10 steps

move 10 steps

move 10 steps

move 10 steps

move 10 steps

move 10 steps

move 10 steps

move 10 steps

move 10 steps

move 10 steps

10 Blue / Motion blocks. Change the distance and you edit all ten.

EFFICIENT / EFICIENTE

repeat (10)

move 10 steps

2 blocks. Change the number once and the whole behavior changes.

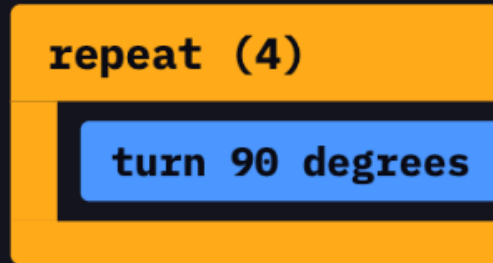
Analogy: a chorus in a song is written once and sung many times.

TEACHER CALLOUT Students often say the loop is "faster." It is not faster on screen – it is easier to read and change. Use those exact words.

03 repeat (N) vs. forever

Both are Orange / Control C-blocks. The difference is how they stop.

COUNT-CONTROLLED



EN: Do the inside blocks a set number of times, then stop.

ES: Repite los bloques de adentro un número fijo de veces y luego se detiene.

Analogy: 10 jumping jacks. You count, you finish.

INFINITE



EN: Do the inside blocks over and over until the program is stopped.

ES: Repite los bloques sin parar hasta que se detenga el programa.

Analogy: your heartbeat. Nobody counts it; it just keeps going.

COMMON PITFALL The forever block has a flat bottom – no blocks can attach under it. Anything students stack below it will never run.

04 repeat until <condition>

A condition-controlled loop: Orange / Control C-block + Cyan / Sensing hexagon.

THE BLOCK



EN: Keep repeating the inside blocks until the condition becomes true, then stop.

ES: Repite los bloques hasta que la condición sea verdadera, y entonces se detiene.

HEXAGON RULE

Hexagon = Boolean = true or false only. It fits only in hexagon holes. / Hexágono = verdadero o falso.

ANALOGY

Walk forward until you reach the wall. You do not count the steps – you check the wall.

Repeat = count the steps. Repeat until = watch for the signal.

COMMON PITFALL If nothing inside the loop can ever make the condition true, the sprite never stops. Ask: "What changes in here?"

05 if <> then vs. if <> then ... else

Decisions, not repetition. One branch or two. / Decisiones: un camino o dos.

SINGLE BRANCH

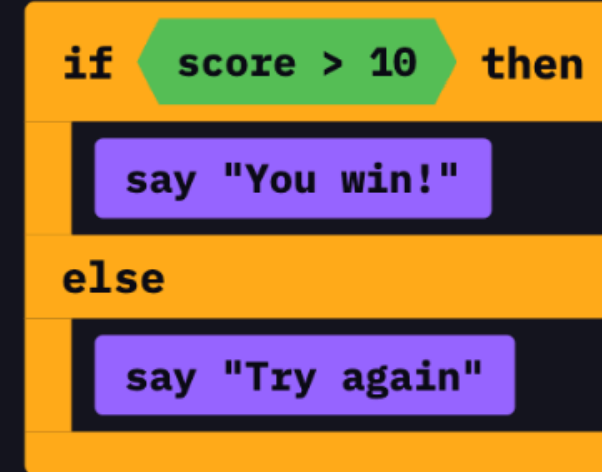


EN: If it is true, run the inside blocks. If it is false, skip them.

ES: Si es verdadero, ejecuta los bloques; si es falso, los salta.

Analogy: if it is raining, take an umbrella. Otherwise, do nothing different.

DUAL BRANCH



EN: If it is true, run the top blocks. If false, run the bottom blocks.

ES: Si es verdadero, ejecuta lo de arriba; si no, lo de abajo.

Analogy: a fork in the road. You take exactly one path.

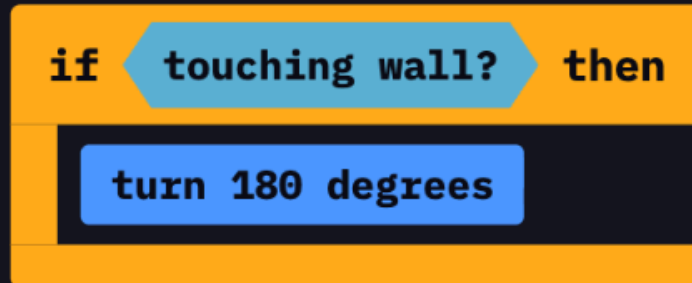
COMMON PITFALL An if checks once and moves on. Students expect it to keep watching – that is what slide 6 fixes.

06

Combining Them: The Continuous Listener

Golden Rule: an **if** inside a **forever** keeps checking. / La regla de oro.

CHECKS ONCE



The sprite asks the question one time, at the start, and never again.

CHECKS ALWAYS



The loop re-asks the question every frame, so the sprite reacts the moment it hits the wall.

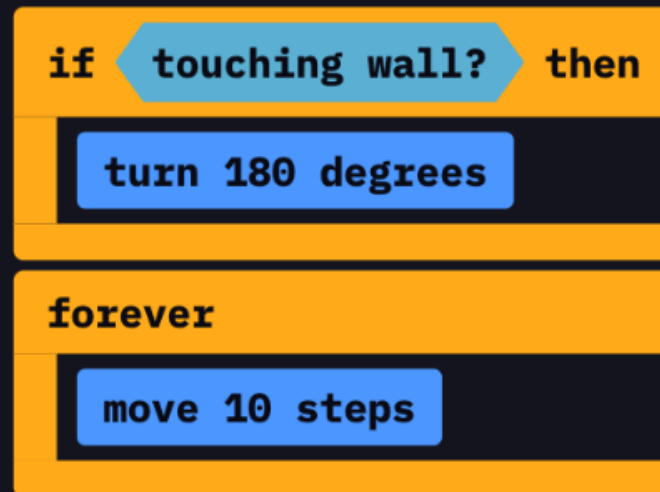
TEACHER CALLOUT Say it as a sentence students can repeat: "A question worth asking twice goes inside a forever loop."

07 Whiteboard Bug Clinic

In pairs: find the bug, then fix it on the board. / En parejas: encuentra el error y arréglalo.

BUG 1

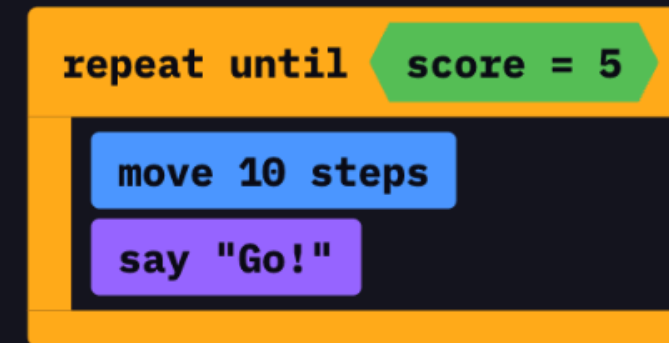
The sprite should bounce every time it hits the wall. It only bounces at the start.



Which block is in the wrong place? / ¿Qué bloque está mal ubicado?

BUG 2

This loop is supposed to stop at 5 points. It never stops.



What is missing inside the loop? / ¿Qué falta dentro del bucle?

EXIT TICKET Name one thing you would build with a forever loop.

Nombra algo que harías con un bucle infinito.